

KHALED NABIL SALAMA

professor

基本信息

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学科领域

生物医学工程

微电子科学与工程

集成电路设计与集成系统

电子与计算机工程

纳米材料与技术

人工智能

材料科学与工程

材料化学

功能材料

生物技术

+2

研究领域

生物医学传感与检测技术

微电子器件与制造

深度学习与神经网络

功能材料与智能材料

纳米合成与组装技术

医用生物材料与植入器械

低功耗设计与能耗优化

生物检测与诊断技术

低功耗电路与能量管理

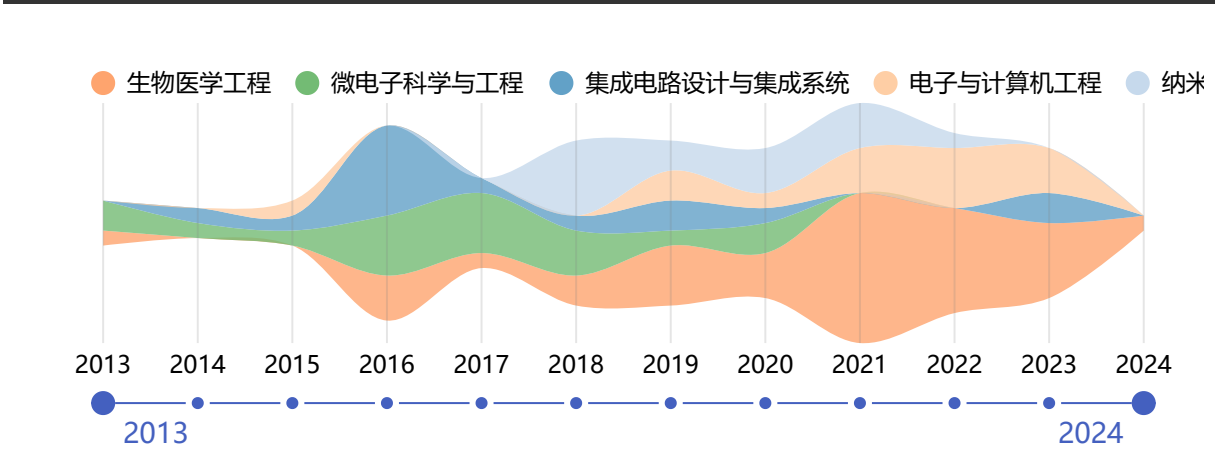
生物信号处理与模式识别

+8

个人简介

Khaled Nabil Salama is a Professor in the Electrical and Computer Engineering and Bioengineering departments at King Abdullah University of Science and Technology (KAUST). He holds a Ph.D. in Electrical Engineering from Stanford University. His research focuses on electronic circuit design, semiconductor fabrication, and the development of analytical platforms for industrial, environmental, and biomedical applications, including neuromorphic circuits for brain emulation. He has published extensively in top-tier journals, contributing to advancements in biosensor technology and wearable biosensor systems. Contact: khaled.salama@kaust.edu.sa

近年发表的论文



Assistive Magnetic Skin System for Speech Reconstruction: An Empowering Technology for Aphonic Individuals
ADVANCED INTELLIGENT SYSTEMS
Ramirez-De Angel, Montserrat; Almansouri, Abdullah S.; Salama, Khaled N.
被引用次数: 0 2024

Enhanced linearity through high-order antisymmetric vibration for MEMS DC power sensor
APPLIED PHYSICS LETTERS
Zou, Xuecui; Jaber, Nizar; Bukhamsin, Abdullah; Yaqoob, Usman; Salama, Khaled Nabil; Fariborzi, Hossein
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Effect of combining UV-C irradiation and vacuum sealing on the shelf life of fresh strawberries and tomatoes
JOURNAL OF FOOD SCIENCE
Damdam, Asrar; Al-Zahrani, Ashwaq; Salah, Lama; Salama, Khaled Nabil
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Smart Multiplex Point-of-Care Platform for Simultaneous Drug Monitoring
ACS APPLIED MATERIALS & INTERFACES
Beduk, Duygu; Beduk, Tutku; de Oliveira Filho, Joseilton; Lahcen, Abdellatif Ait; Aldemir, Ebru; Celik, Emine Guler; Salama, Khaled Nabil; Timur, Suna
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Iron Single-Atom Catalysts on MXenes for Ultrasensitive Monitoring of Adrenal Tumor Markers and Cellular Dopamine
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Shetty, Saptami Suresh; El-Demellawi, Jehad K.; Khan, Yusuf; Hedhili, Mohamed N.; Arul, P.; Mani, Veerappan; Alshareef, Husam N.; Salama, Khaled Nabil
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Fabrication and Characterization of Porous Microneedles for Enhanced Fluid Injection and Suction: A Two-Photon Polymerization Approach

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Resistive Neural Hardware Accelerators

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Empowering Electrochemical Biosensors with AI: Overcoming Interference for Precise Dopamine Detection in Complex Samples

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de Oliveira Filho, Jose Ilton; Faleiros, Murilo Calil; Ferreira, Daisy Camargo; Mani, Veerappan; Salama, Khaled Nabil

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An interconnect-free micro-electromechanical 7-bit arithmetic device for multi-operand programmable computing

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Architectural Trade-Off Analysis for Accelerating LSTM Network Using Radix-r OBC Scheme

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A Mechanical Memory With Capacitance Modulation of Nonlinear Resonance

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Synergistic multi-source ambient RF and thermal energy harvester for green IoT applications

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Bakytbekov, Azamat; Nguyen, Thang Q.; Zhang, Ge; Strano, Michael S.; Salama, Khaled N.; Shamim, Atif

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Polyoxometalate-cyclodextrin supramolecular entities for real-time in situ monitoring of dopamine released from neuroblastoma cells

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Efficient training of spiking neural networks with temporally-truncated local backpropagation through time

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Self-Powered Flexible Wearable Sensing Platform for Ascorbic Acid Detection in Sweat

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Highly Sensitive Wireless NO₂ Gas Sensing System

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A flexible capacitive photoreceptor for the biomimetic retina

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Self-powered weather station for remote areas and difficult-access locations

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Metal-Organic Frameworks Meet Molecularly Imprinted Polymers: Insights and Prospects for Sensor Applications

ACS APPLIED MATERIALS & INTERFACES

Lahcen, Abdellatif Ait; Surya, Sandeep G.; Beduk, Tutku; Vijjapu, Mani Teja; Lamaoui, Abderrahman; Durmus, Ceren; Timur, Suna; Shekhah, Osama; Mani, Veerappan; Amine, Aziz; Eddaoudi, Mohamed; Salama, Khaled Nabil

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Toward the Optimal Design and FPGA Implementation of Spiking Neural Networks

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A hardware/software co-design methodology for in-memory processors

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JOURNAL OF ELECTROANALYTICAL CHEMISTRY

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A Portable Molecularly Imprinted Sensor for On-Site and Wireless Environmental Bisphenol A Monitoring

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Efficient Neuromorphic Hardware Through Spiking Temporal Online Local Learning

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Graphitic Carbon Nitride and IGZO Bio-FET for Rapid Diagnosis of Myocardial Infarction

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Smartphone-Based Multiplexed Biosensing Tools for Health Monitoring

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Implementation of a DPU-Based Intelligent Thermal Imaging Hardware Accelerator on FPGA

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Multiplexed sensing techniques for cardiovascular disease biomarkers- A review

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Polarization Insensitive and Transparent Frequency Selective Surface for Dual Band GSM Shielding

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Enhanced acoustic pressure sensors based on coherent perfect absorber-laser effect (vol 129, 104902, 2021)

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A highly selective electron affinity facilitated H₂S sensor: the marriage of tris(keto-hydrazone) and an organic field-effect transistor

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Yuvaraja, Saravanan; Bhyranalyar, Veerabhadraswamy Nagarajappa; Bhat, Sachin Ashok; Surya, Sandeep Goud; Yelamaggad, Channabasaveshwar Veerappa; Salama, Khaled Nabil

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Highly Selective Self-Powered Organic-Inorganic Hybrid Heterojunction of a Halide Perovskite and InGaZnO NO₂ Sensor

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Vijjapu, Mani Teja; Surya, Sandeep G.; He, Jr-Hau; Salama, Khaled N.

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Tris(Keto-Hydrazone): A Fully Integrated Highly Stable and Exceptionally Sensitive H₂S Capacitive Sensor

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Binary transition metal oxide modified laser-scribed graphene electrochemical aptasensor for the accurate and sensitive screening of acute myocardial infarction

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Rauf, Sakandar; Mani, Veerappan; Lahcen, Abdellatif Ait; Yuvaraja, Saravanan; Beduk, Tutku; Salama, Khaled N.

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Coating of Conducting and Insulating Threads with Porous MOF Particles through Langmuir-Blodgett Technique

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A Wideband Magnetic Frequency Up-Converter Energy Harvester

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2021

An Assistive Magnetic Skin System: Enabling Technology for Quadriplegics

ADVANCED ENGINEERING MATERIALS

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Fractional-Order Inductor: Design, Simulation, and Implementation

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Hardware Acceleration of High Sensitivity Power-Aware Epileptic Seizure Detection System Using Dynamic Partial Reconfiguration

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Breath as the mirror of our body is the answer really blowing in the wind? Recent technologies in exhaled breath analysis systems as non-invasive sensing platforms

TRAC-TRENDS IN ANALYTICAL CHEMISTRY

Beduk, Tutku; Durmus, Ceren; Hanoglu, Simge Balaban; Beduk, Duygu; Salama, Khaled Nabil; Goksel, Tuncay; Turhan, Kutsal; Timur, Suna

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Electrochemical sensors targeting salivary biomarkers: A comprehensive review

TRAC-TRENDS IN ANALYTICAL CHEMISTRY

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Laser-scribed graphene sensor based on gold nanostructures and molecularly imprinted polymers: Application for Her-2 cancer biomarker detection

SENSORS AND ACTUATORS B-CHEMICAL

Lahcen, Abdellatif Ait; Rauf, Sakandar; Aljedaibi, Abdulrahman; de Oliveira Filho, Jose Ilton; Beduk, Tutku; Mani, Veerappan; Alshareef, Husam N.; Salama, Khaled N.

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Towards a low cost fully integrated IGZO TFT NO₂ detection and quantification: A solution-processed approach

SENSORS AND ACTUATORS B-CHEMICAL

Vijjapu, Mani Teja; Surya, Sandeep; Zalte, Maruti; Yuvaraja, Saravanan; Baghini, Maryam Shojaei; Salama, Khaled N.

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Anti-poisoning electrode for real-time in-situ monitoring of hydrogen sulfide release

SENSORS AND ACTUATORS B-CHEMICAL

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Rapid Point-of-Care COVID-19 Diagnosis with a Gold-Nanoarchitecture-Assisted Laser-Scribed Graphene Biosensor

ANALYTICAL CHEMISTRY

Beduk, Tutku; Beduk, Duygu; de Oliveira Filho, Jose Ilton; Zihnioglu, Figen; Cicek, Candan; Sertoz, Ruchan; Arda, Bilgin; Goksel, Tuncay; Turhan, Kutsal; Salama, Khaled N.; Timur, Suna

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Inherent Surface Activation of Laser-Scribed Graphene Decorated with Au and Ag Nanoparticles: Simultaneous Electrochemical Behavior toward Uric Acid and Dopamine

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IMCA: An Efficient In-Memory Convolution Accelerator

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Yantir, Hasan Erdem; Eltawil, Ahmed M.; Salama, Khaled N.

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Gold nanostructured laser-scribed graphene: A new electrochemical biosensing platform for potential point-of-care testing of disease biomarkers

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Rauf, Sakandar; Lahcen, Abdellatif Ait; Aljedaibi, Abdulrahman; Beduk, Tutku; de Oliveira Filho, Jose Ilton; Salama, Khaled N.

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Neural Coding in Spiking Neural Networks: A Comparative Study for Robust Neuromorphic Systems

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Guo, Wenzhe; Fouda, Mohammed E.; Eltawil, Ahmed M.; Salama, Khaled Nabil

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Enhanced Graphene Sensors via Multi-Lasing Fabrication

IEEE SENSORS JOURNAL

Kaidarova, Altynay; Vijjapu, Mani Teja; Telegenov, Kuat; Przybysz, Alexander; Salama, Khaled Nabil; Kosel, Jurgan

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Blue energy fuels: converting ocean wave energy to carbon-based liquid fuels via CO₂ reduction

ENERGY & ENVIRONMENTAL SCIENCE

Leung, Siu-Fung; Fu, Hui-Chun; Zhang, Maolin; Hassan, Ali H.; Jiang, Tao; Salama, Khaled N.; Wang, Zhonglin; He, Jr-Hau

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A Compact, Passive Frequency-Hopping Harmonic Sensor Based on a Microfluidic Reconfigurable Dual-Band Antenna

Fully Integrated Organic Field-Effect Transistor Platform to Detect and to Quantify NO₂ Gas

PHYSICA STATUS SOLIDI-RAPID RESEARCH LETTERS

Yuvaraja, Saravanan; Surya, Sandeep G.; Vijjapu, Mani Teja; Chernikova, Valeriya; Shekhah, Osama; Eddaoudi, Mohamed; Salama, Khaled Nabil

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2020

Selective Toluene Detection with Mo₂CTx MXene at Room Temperature

ACS APPLIED MATERIALS & INTERFACES

Guo, Wenzhe; Surya, Sandeep G.; Babar, Vasudeo; Ming, Fangwang; Sharma, Sitansh; Alshareef, Husam N.; Schwingenschlogl, Udo; Salama, Khaled N.

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2020

Highly Selective Metal-Organic Framework Textile Humidity Sensor

ACS APPLIED MATERIALS & INTERFACES

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Realization of an Ultrasensitive and Highly Selective OFET NO₂ Sensor: The Synergistic Combination of PDVT-10 Polymer and Porphyrin-MOF

ACS APPLIED MATERIALS & INTERFACES

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2020

Methanol and Humidity Capacitive Sensors Based on Thin Films of MOF Nanoparticles

ACS APPLIED MATERIALS & INTERFACES

Andres, Miguel A.; Vijjapu, Mani Teja; Surya, Sandeep G.; Shekhah, Osama; Salama, Khaled Nabil; Serre, Christian; Eddaoudi, Mohamed; Roubeau, Olivier; Gascon, Ignacio

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Fully Transparent Transceiver Using Single Binary Oxide Thin Film Transistors

ADVANCED ELECTRONIC MATERIALS

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Organic field-effect transistor-based flexible sensors

CHEMICAL SOCIETY REVIEWS

Yuvaraja, Saravanan; Nawaz, Ali; Liu, Qian; Dubal, Deepak; Surya, Sandeep G.; Salama, Khaled N.; Sonar, Prashant

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Fully Integrated Indium Gallium Zinc Oxide NO₂ Gas Detector

ACS SENSORS

Vijjapu, Mani Teja; Surya, Sandeep G.; Yuvaraja, Saravanan; Zhang, Xixiang; Alshareef, Husam N.; Salama, Khaled N.

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2020

An in -field integrated capacitive sensor for rapid detection and quantification of soil moisture

SENSORS AND ACTUATORS B-CHEMICAL

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One-step electrosynthesized molecularly imprinted polymer on laser scribed graphene bisphenol a sensor

SENSORS AND ACTUATORS B-CHEMICAL

Beduk, Tutku; Lahcen, Abdellatif Ait; Tashkandi, Nouran; Salama, Khaled Nabil

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A paper-based inkjet-printed PEDOT:PSS/ZnO sol-gel hydrazine sensor

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Beduk, Tutku; Bihar, Eloise; Surya, Sandeep G.; Castillo, Aminta N.; Inal, Sahika; Salama, Khaled N.

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A Dual-Mode Nested Rectifier for Ambient Wireless Powering in CMOS Technology

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Almansouri, Abdullah S.; Kosel, Jurgen; Salama, Khaled N.

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Multi-source ambient energy harvester based on RF and thermal energy: Design, testing, and IoT application

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Efficient Acceleration of Stencil Applications through In-Memory Computing

MICROMACHINES

Yantir, Hasan Erdem; Eltawil, Ahmed M.; Salama, Khaled N.

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Laser scribed graphene: A novel platform for highly sensitive detection of electroactive biomolecules

BIOSENSORS & BIOELECTRONICS

Ghanam, Abdelghani; Lahcen, Abdellatif Ait; Beduk, Tutku; Alshareef, Husam N.; Amine, Aziz; Salama, Khaled Nabil

Electrochemical sensors and biosensors using laser-derived graphene: A comprehensive review

BIOSENSORS & BIOELECTRONICS

Lahcen, Abdellatif Ait; Rauf, Sakandar; Beduk, Tutku; Durmus, Ceren; Aljedaibi, Abdulrahman; Timur, Suna; Alshareef, Husam N.; Amine, Aziz; Wolfbeis, Otto S.; Salama, Khaled N.

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Growth of large-scale MoS₂ nanosheets on double layered ZnCo₂O₄ for real-time in situ H₂S monitoring in live cells

JOURNAL OF MATERIALS CHEMISTRY B

Mani, Veerappan; Selvaraj, Shanthi; Jeromiyas, Nithiya; Huang, Sheng-Tung; Ikeda, Hiroya; Hayakawa, Yasuhiro; Ponnusamy, Suru; Muthamizhchelvan, Chellamuthu; Salama, Khaled Nabil

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2020

A label-free aptasensor FET based on Au nanoparticle decorated Co₃O₄ nanorods and a SWCNT layer for detection of cardiac troponin T protein

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Surya, Sandeep G.; Majhi, Sanjit M.; Agarwal, Dilip K.; Lahcen, Abdellatif Ait; Yuvaraja, Saravanan; Chappanda, Karumbaiah N.; Salama, Khaled N.

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Unsupervised Adaptive Weight Pruning for Energy-Efficient Neuromorphic Systems

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Spectrometer-Free Graphene Plasmonics Based Refractive Index Sensor

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Zhang, Li; Farhat, Mohamed; Salama, Khaled Nabil

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Photo-carrier extraction by triboelectricity for carrier transport layer-free photodetectors

NANO ENERGY

Hsiao, Vincent K. S.; Leung, Siu-Fung; Hsiao, Yung-Chi; Kung, Po-Kai; Lai, Ying-Chih; Lin, Zong-Hong; Salama, Khaled N.; Alshareef, Husam N.; Wang, Zhong Lin; He, Jr-Hau

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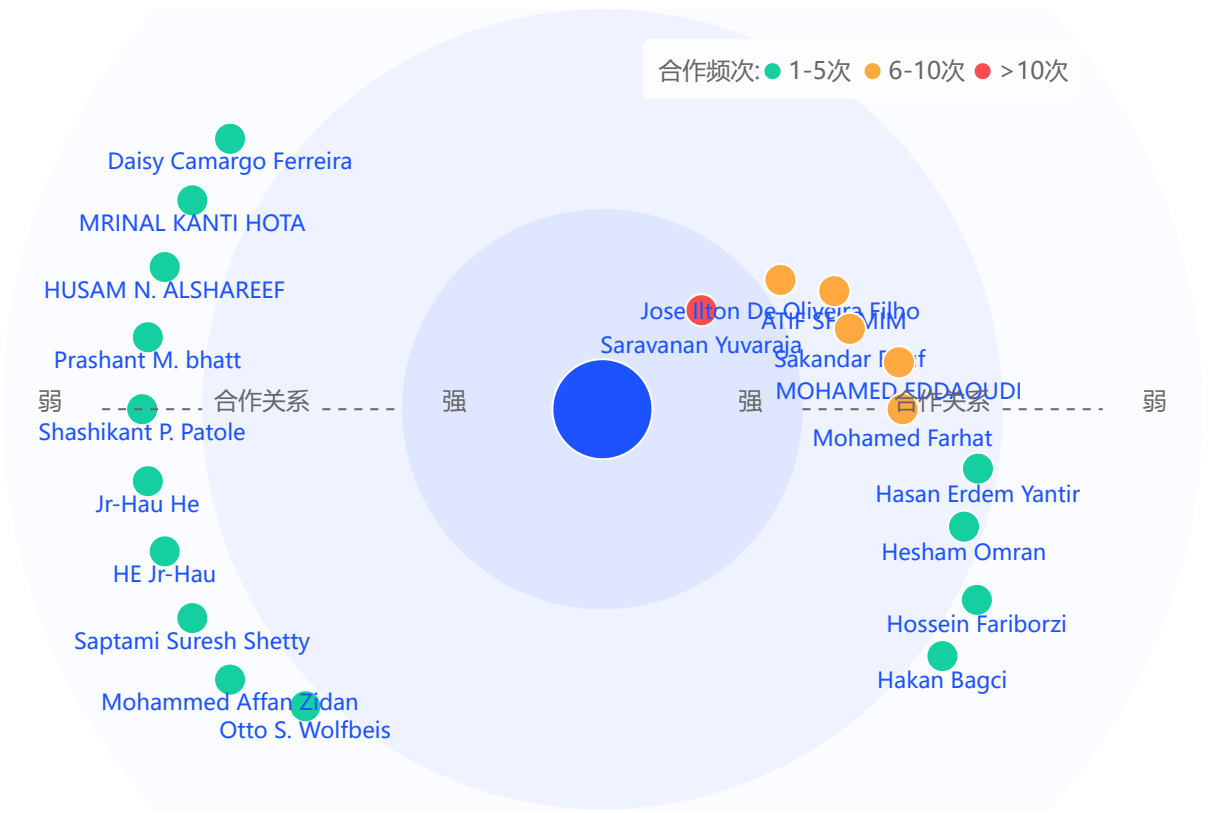
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